

TEAM LINEAR

EEE 4308 Project
Phase A,B,C, D and E (HW)

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INTRODUCTION:

This report documents the development of PCB designs for phase A, B, C, D and E of the hardware phase of the project for EEE4308. The schematics were created in the software phases for this project. This also describes some incremental changes made to the schematics created previously.

The Game

A two-player Snake-Ladder Game is to be made using electronic circuits.

The board will be a 4x4 grid, counting from 0 to 15. Each square will have 4 LED indicators.

At the start of the game, both players will be located at the square 0, which will be indicated by their respective LED indicators. A random number generator (the dice) will generate the numbers 1,2,3 randomly.

Players will take turn rolling the dice. The player rolling the will advance by n position, where n is the output of the dice.

There will be special positions on the board called Ladders and Snakes, each having two ends. Upon landing on the start point of the ladder, the player will be promoted to the other end of the ladder. In the case of the snake, the player will be demoted. The starting positions and lengths of these will be specified by the user at the start of the game.

The player reaching the position 15 on the board first will win.

REQUIREMENTS:

For this phase, we were required to design PCBs for the whole project (A, B, C, D and E).

OVERVIEW

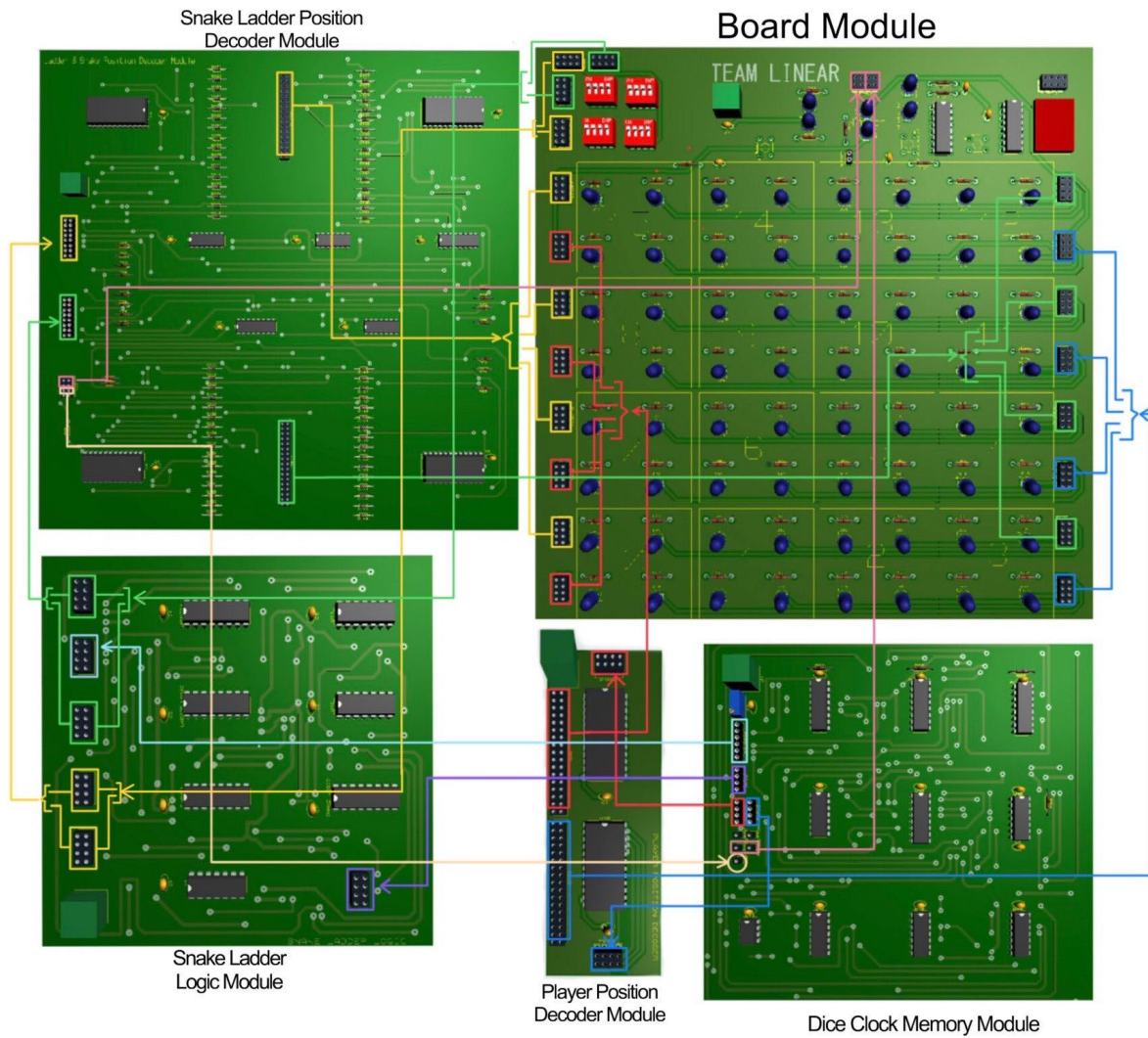


Figure a: Connection Schema

BOARD MODULE

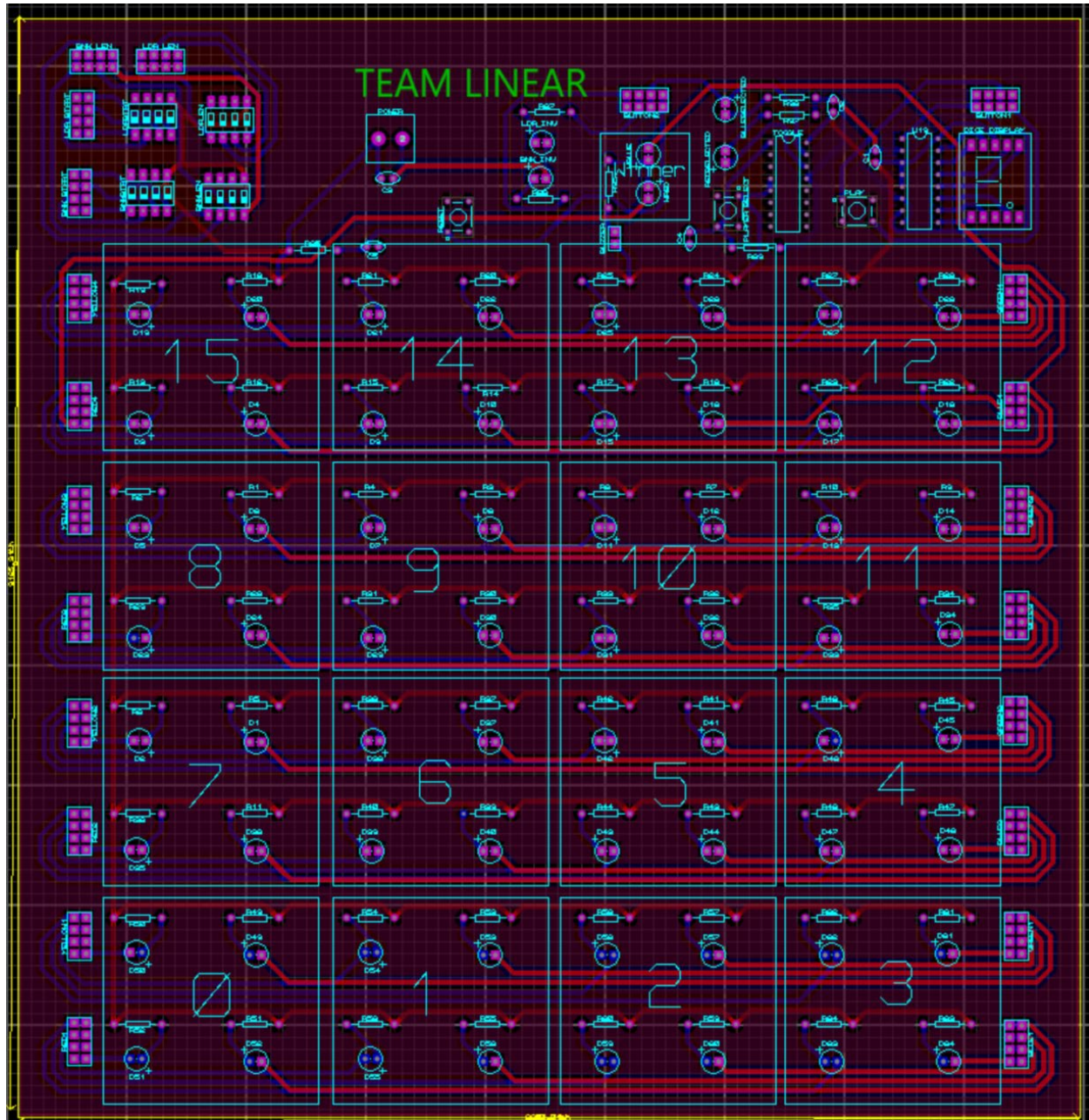


Figure b: PCB design view of Board Module: Top Copper

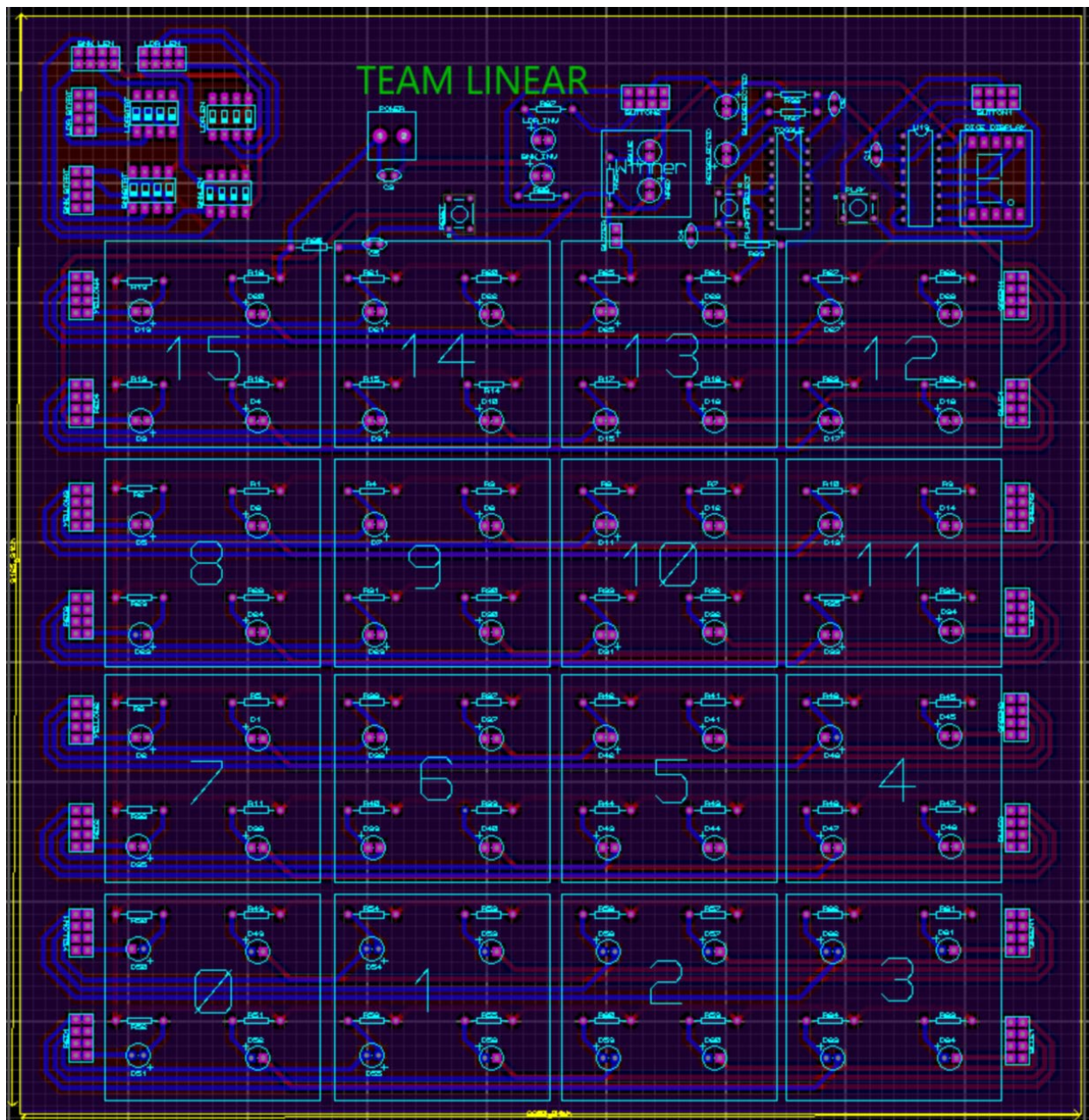


Figure c:PCB design view of Board Module: Bottom Copper

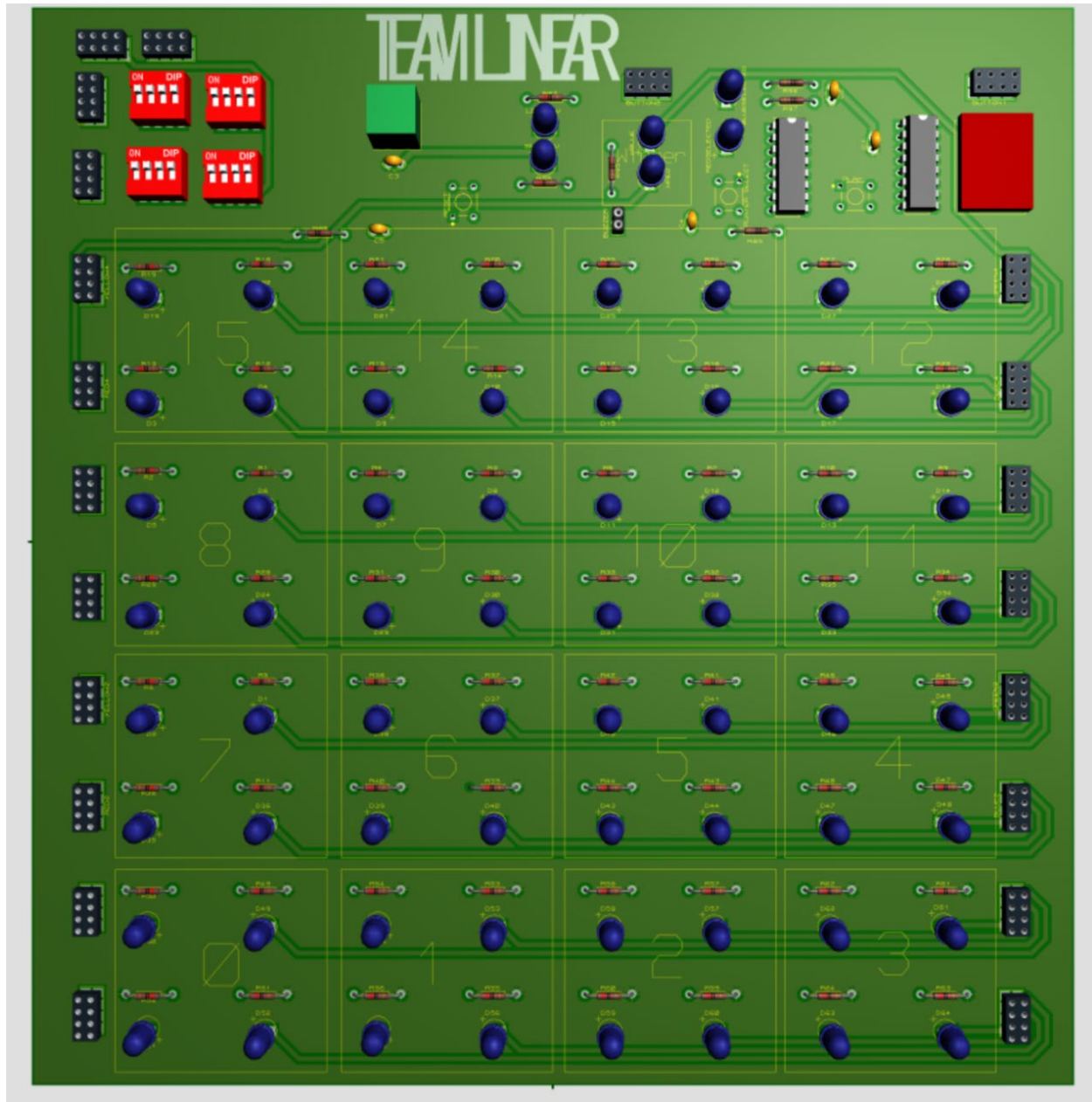


Figure d: 3D render of Board Module: Top view

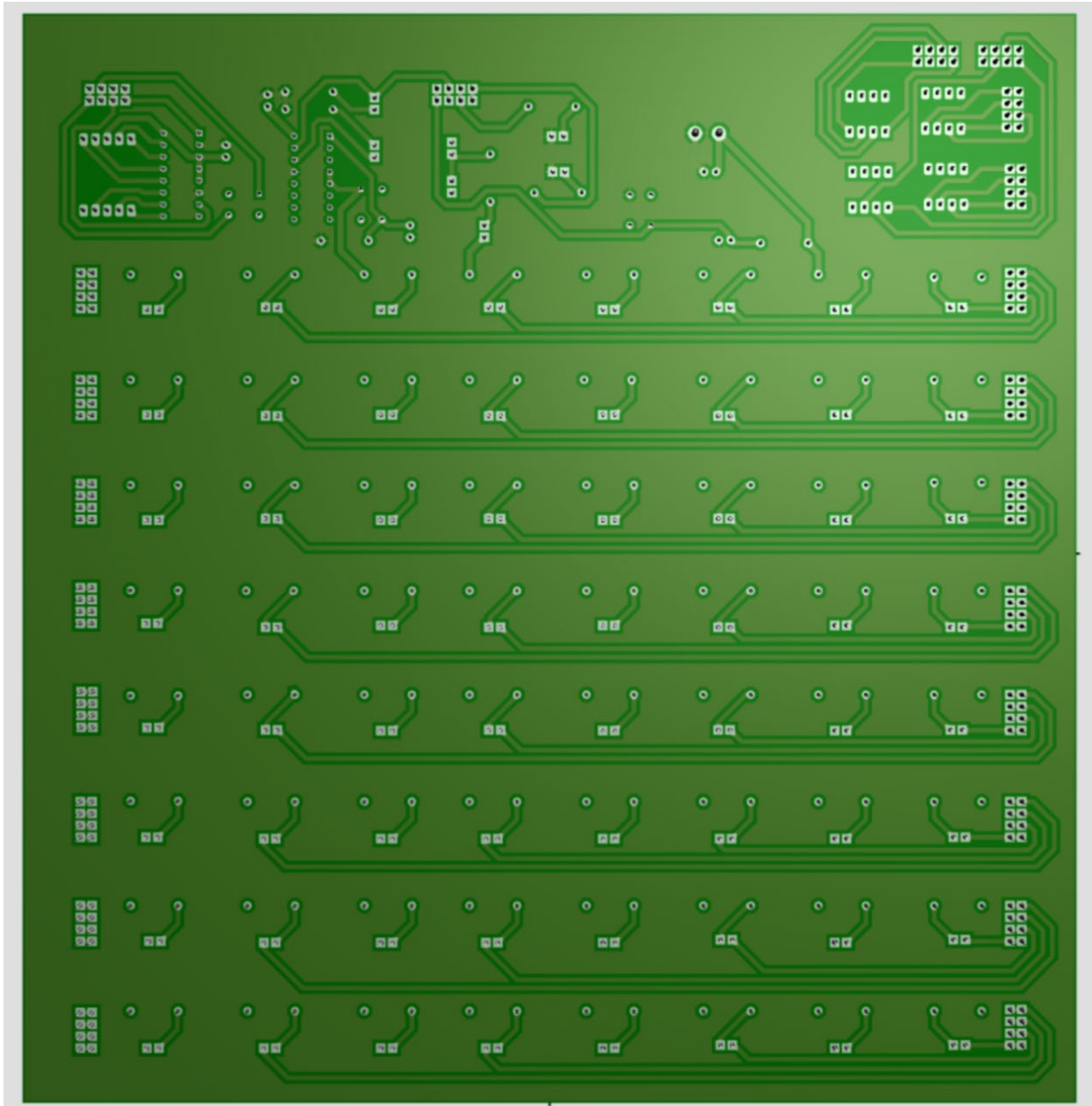


Figure e: 3D render of Board Module: Bottom View

This board performs the **functionalities** of:

- Board Module
- Winner Module

And the **interfaces** for:

- Reset module
- Player Selector Module
- Dice Module

The design is 8.85×9.125 sq. inch. It has no vias. Ratsnest (power plane) has been configured with Top Layer for VCC and Bottom Layer for Ground. All the signal traces are TH40, and the power traces are TH50.

SNAKE-LADDER POSITION DECODER

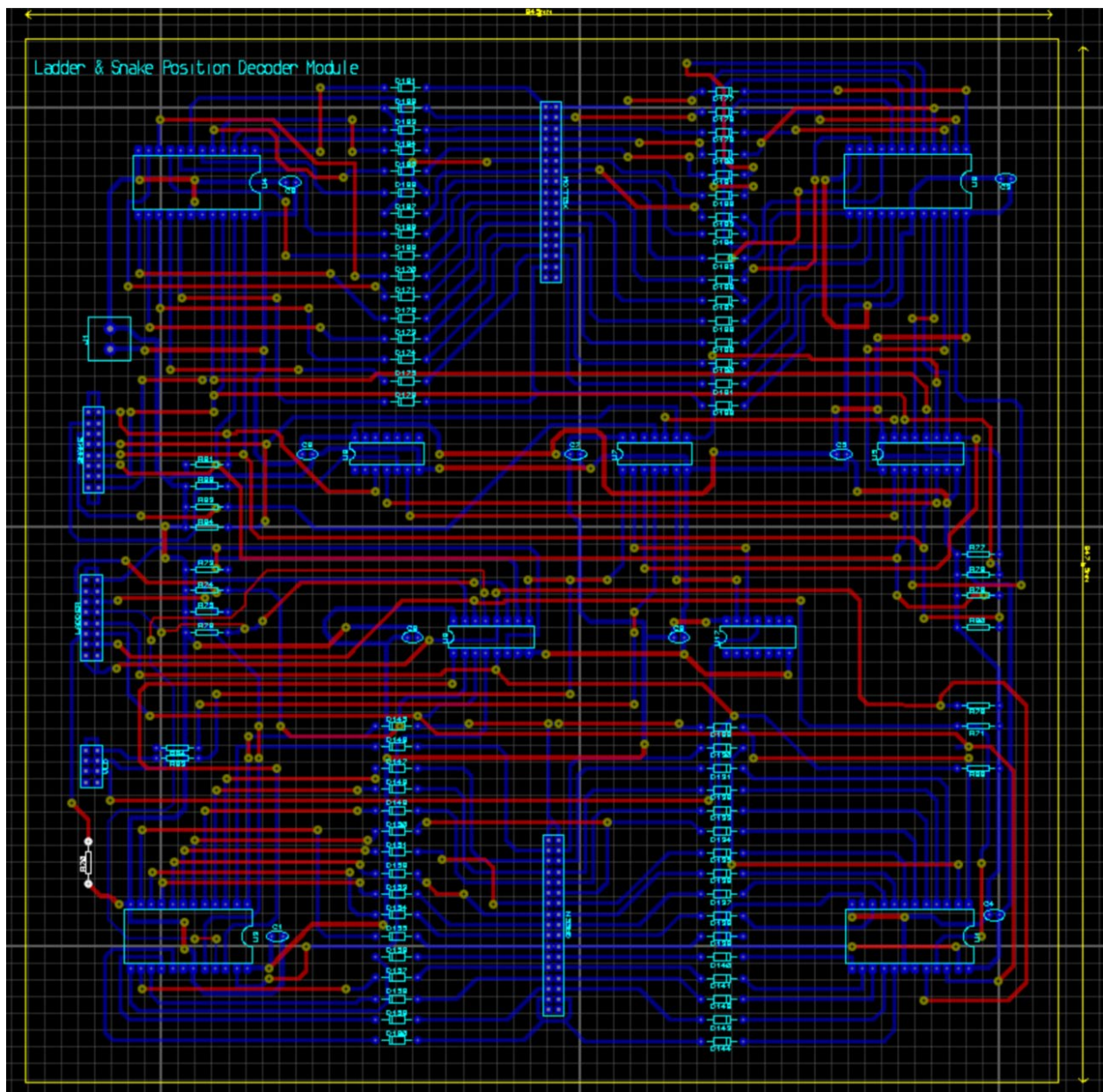


Figure f: PCB design view of Snake-Ladder Position Decoder

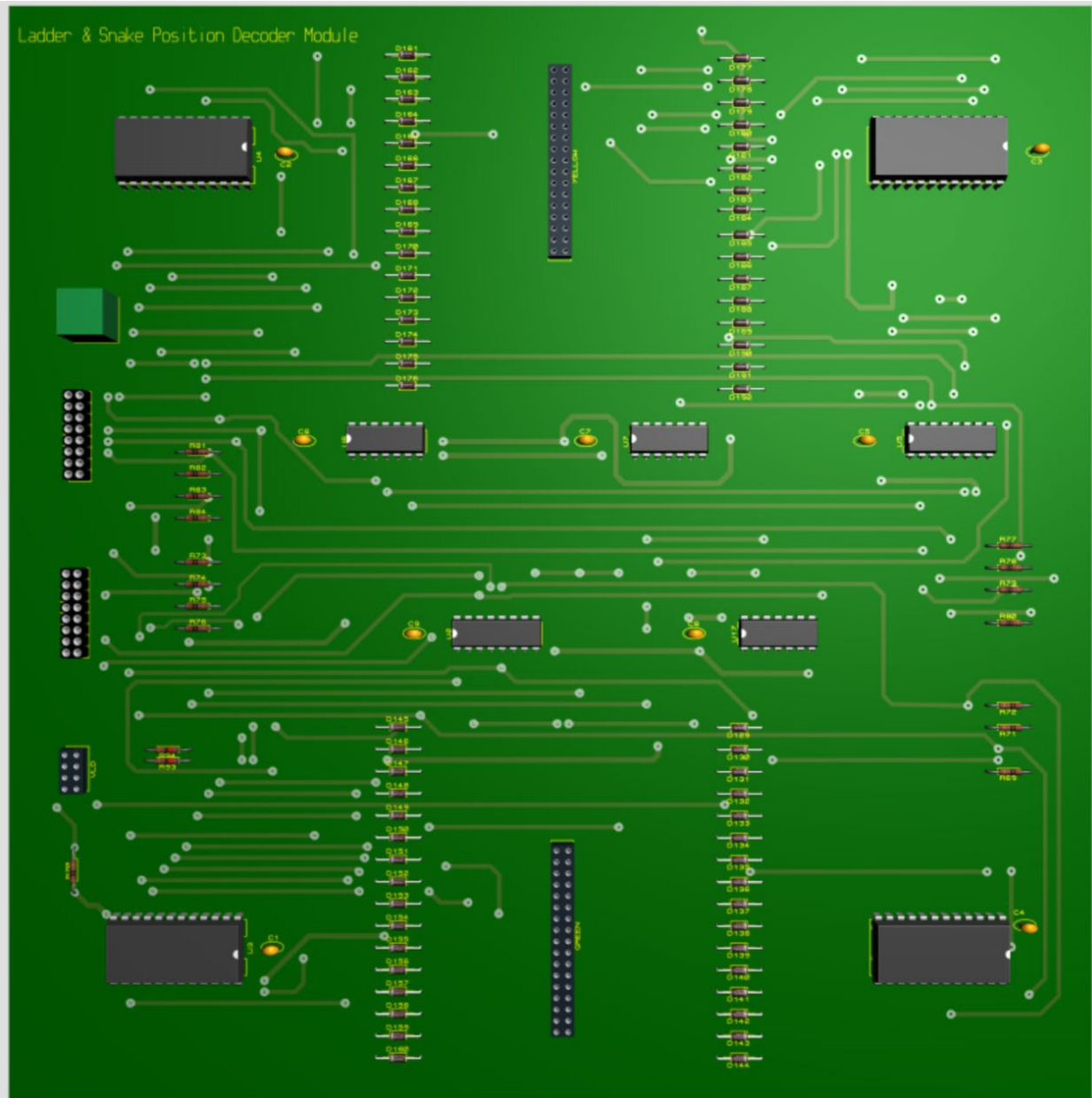


Figure g: 3D render of Snake-Ladder Position Decoder: Top view

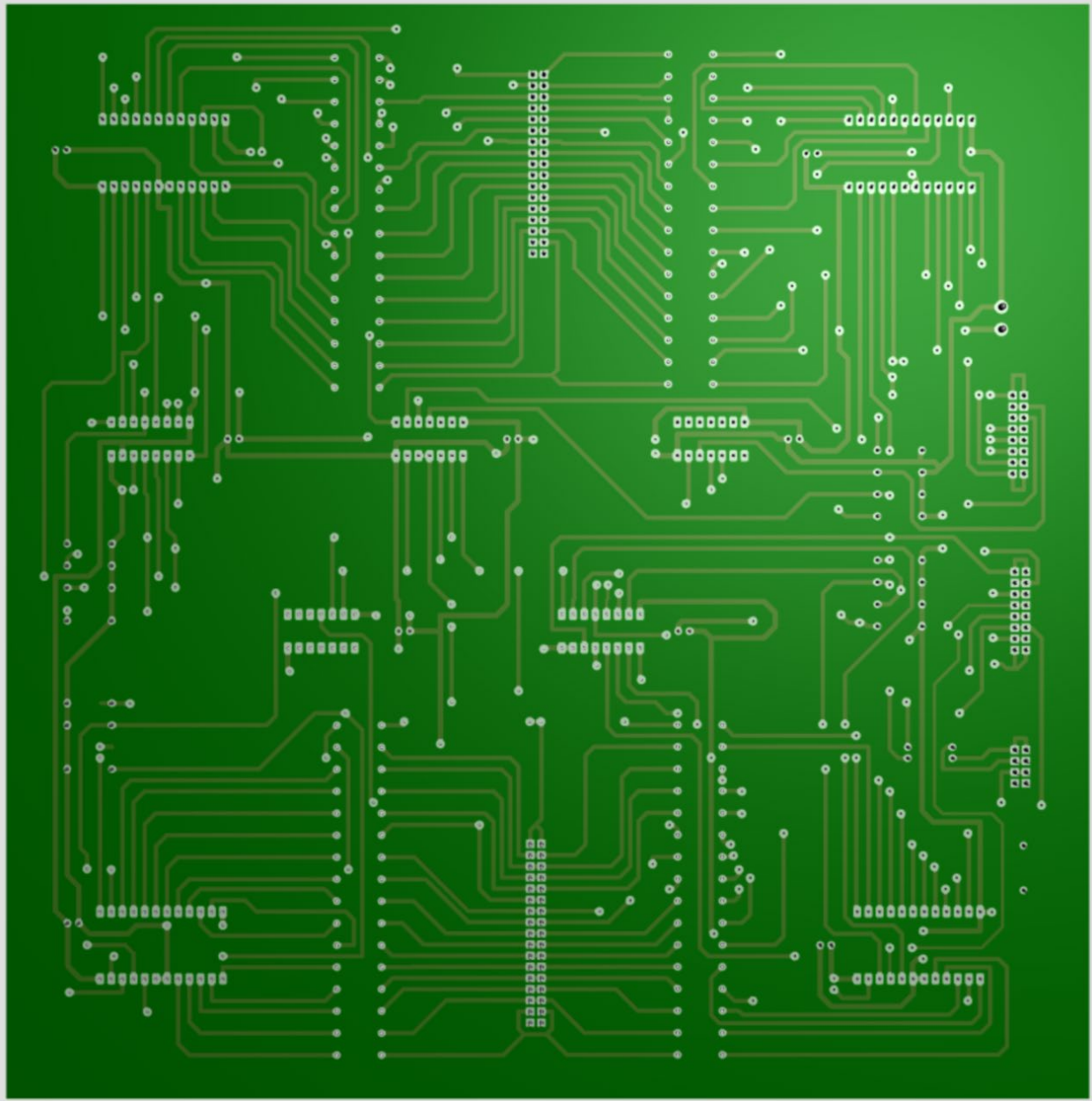


Figure h: 3D render of Snake-Ladder Position Decoder: Bottom view

This board includes the functionalities of Snake-Ladder Position decoder module. It is to be connected with:

- Board module PCB
- Snake-Ladder Logic PCB

Its dimensions are 9.645×9.744 sq. inch.

SNAKE-LADDER LOGIC MODULE

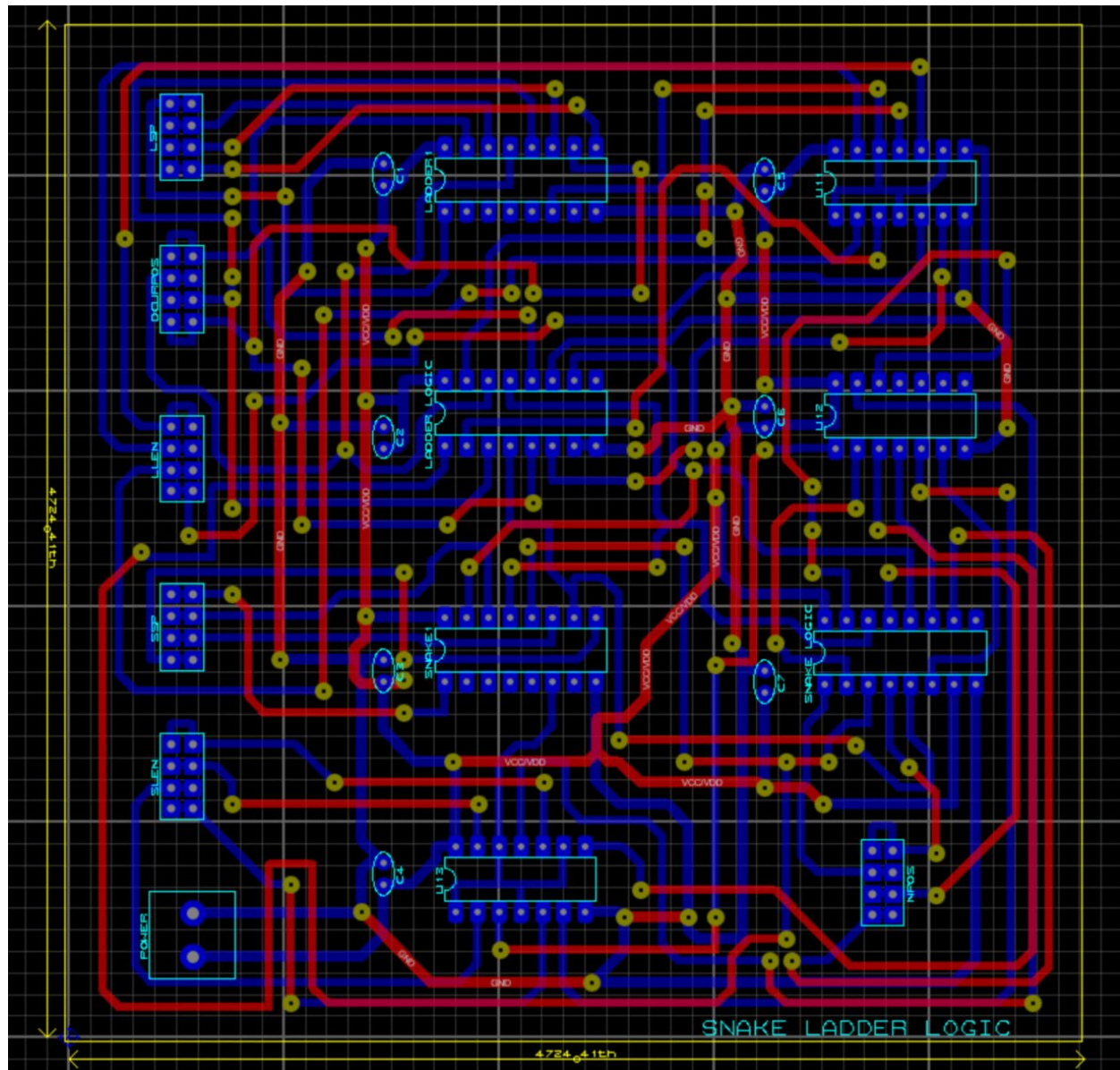


Figure i- PCB design view of Snake-Ladder Logic Module

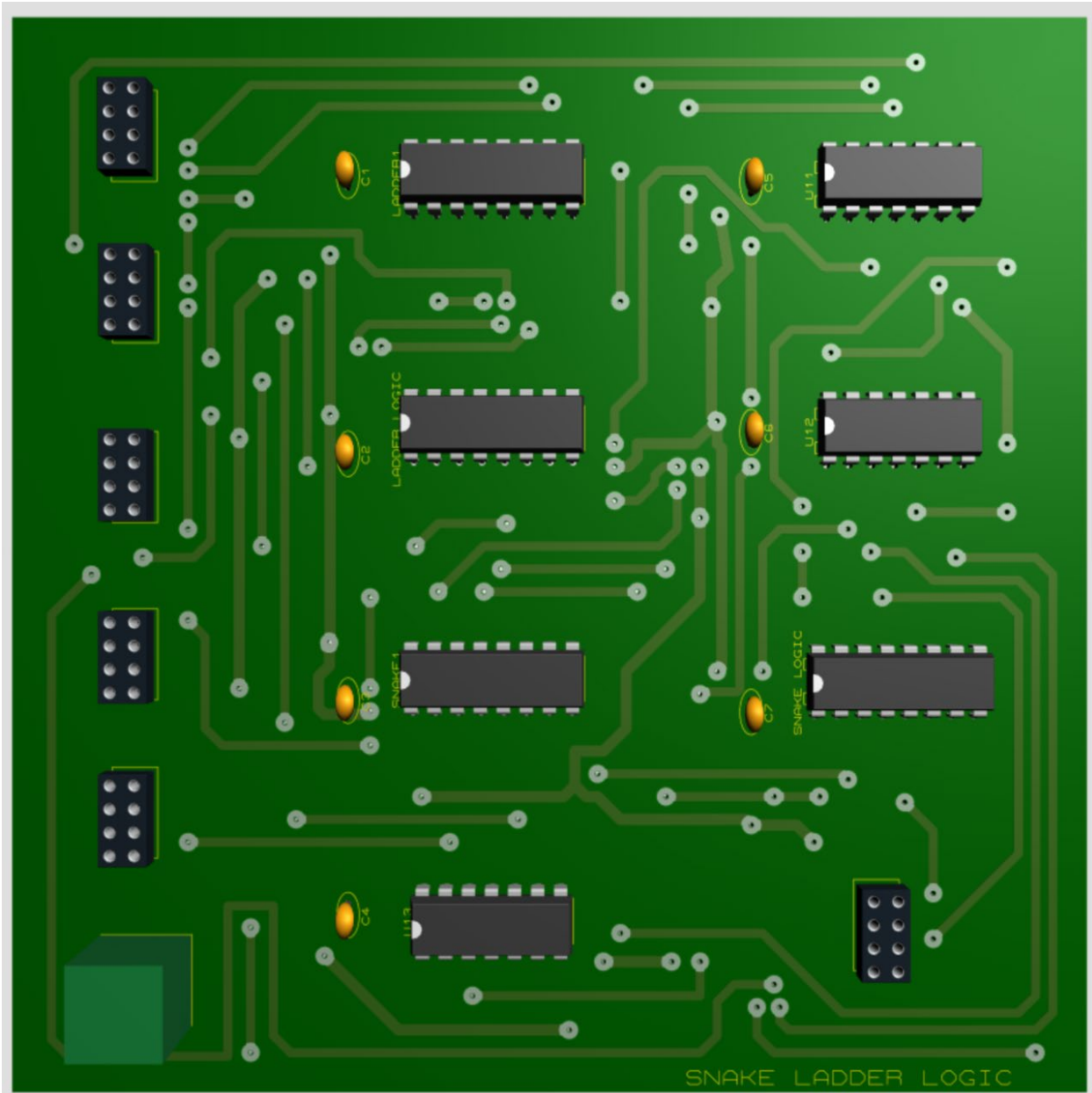


Figure j:3D Render of Snake-Ladder Logic: Top View

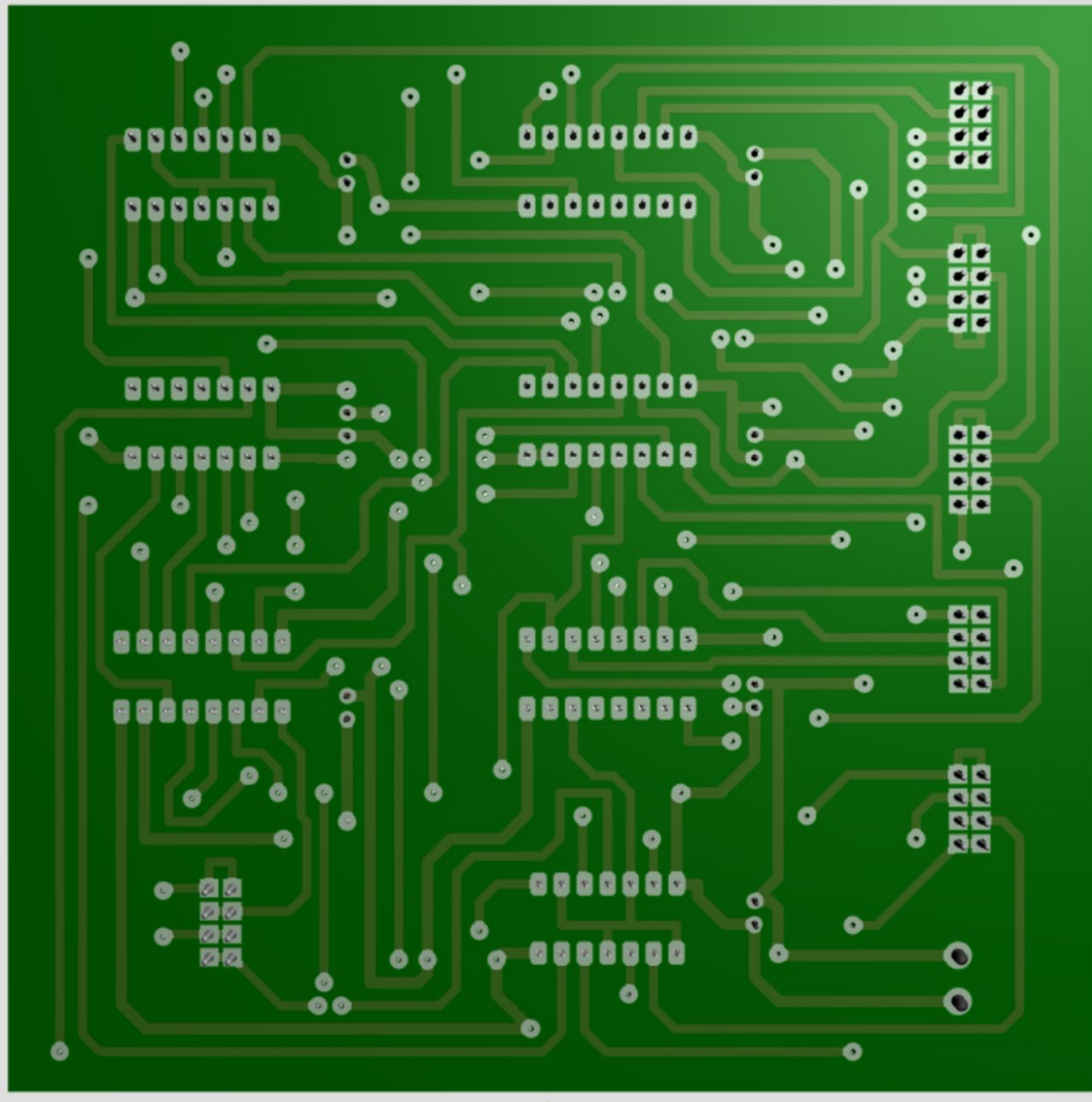


Figure k: 3D render of Snake-Ladder Logic: Bottom View

The dimensions of this board are 4.724×4.724 sq inch. It performs the functionalities of the snake-ladder logic module.

It is to be connected with:

- Snake-ladder position decoder board/ Board module
- Memory Module

DICE CLOCK AND MEMORY MODULE

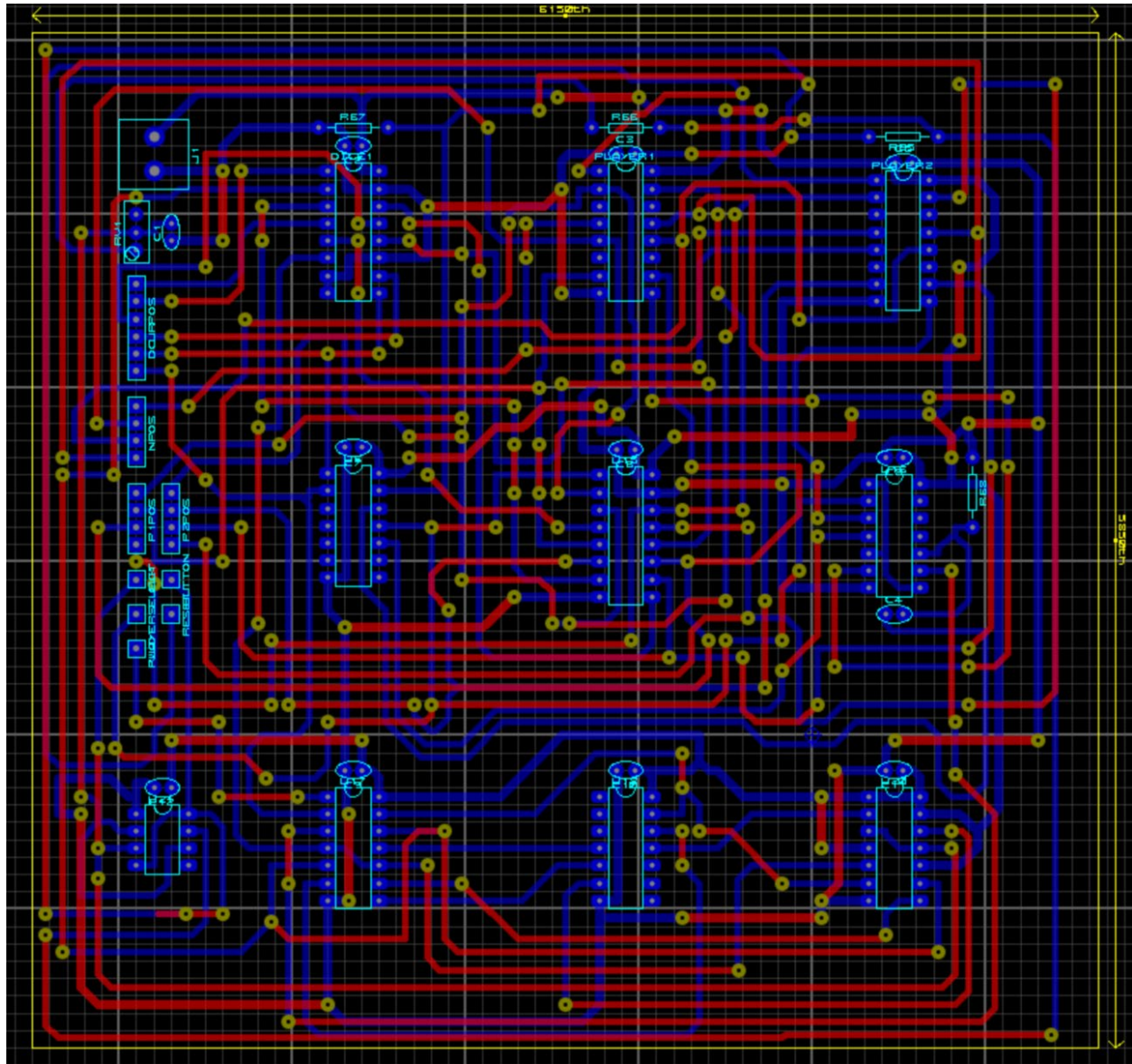


Figure 1: PCB design view of Dice Clock and Memory Module

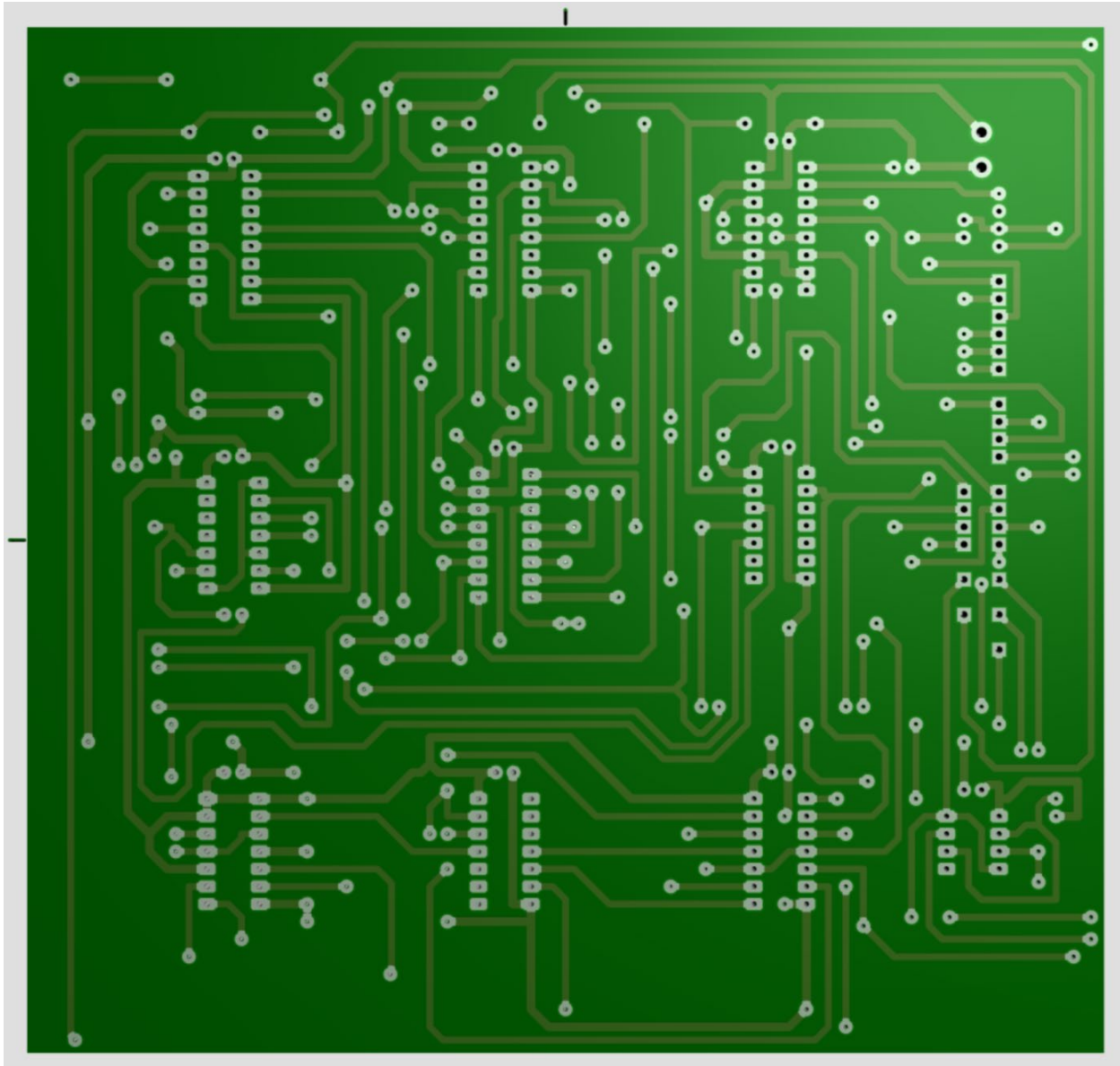


Figure n: 3D render of Dice Clock and Memory Module: Bottom view

This board has the dimensions of 5.2×5.9 sq. inch. This board performs the functionalities of:

- Dice Module
- Clock Module
- Memory Module
- Player Selector Module
- Reset Module

It is to be connected to:

- Board module PCB

- Snake-Ladder Logic PCB
- Player Position Decoder PCB

PLAYER POSITION DECODER

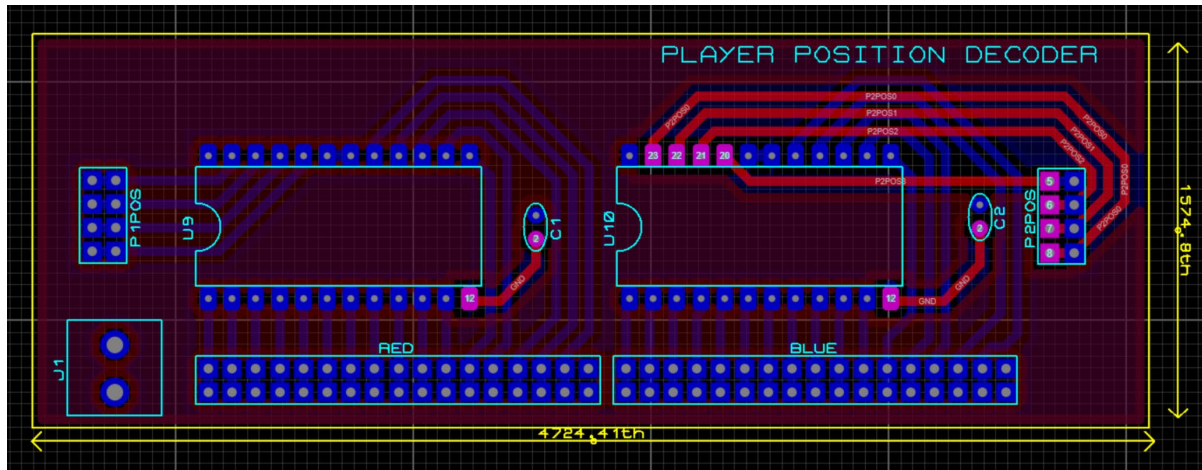


Figure o: PCB design view of Player Position Decoder

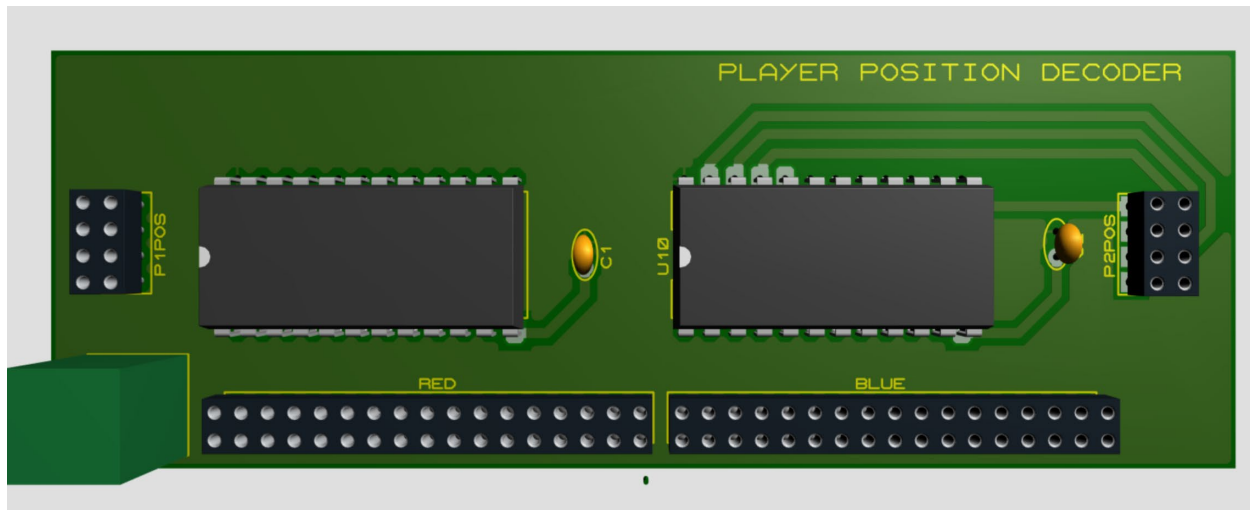


Figure p: 3D render of Player Position Decoder: Top View

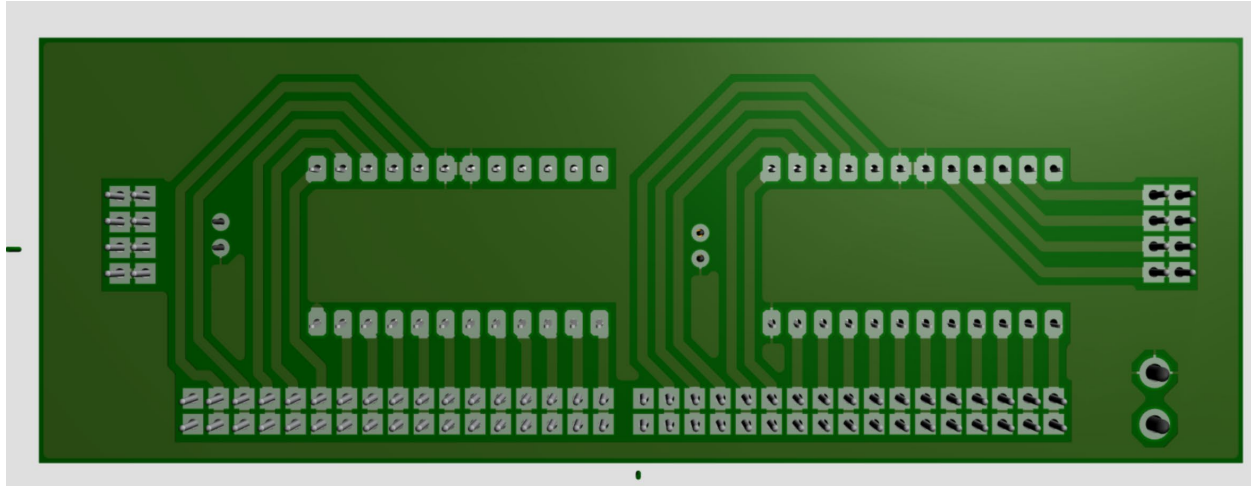


Figure q: 3D render of Player Position Decoder: Bottom View

This board performs the functionalities of player position decoder module. Its dimensions are 1.6×4.7 sq. inch. Ratsnest (power plane) has been added to the top and bottom layer for GND

MEMBER CONTRIBUTIONS

220021202- K.M. Tausif Ahnaf

- Designed a preliminary version of the Board Module PCB by placing the components

220021208- Arham Takib

- Created the copper trace pdfs that is required for printing
- Provided suggestions from time to time regarding design choices

220021242- Mubtasim Labib

- Provided clear instructions to other members to ensure a proper collaborative workflow
- Designed the Snake-Ladder Logic PCB
- Fixed the placement errors in Tausif's Board Module
- Designed the Snake-Ladder Position decoder by tweaking the designs Tashfeea had provided
- Enhanced the Player Position decoder by adding Ratsnest (Power plane) to Tashfeea's Design
- Provided suggestions from time to time regarding design choices

220021248- Tanvir Hossain

- Designed the combined PCB for Dice, Clock and Memory Module
- Provided suggestions from time to time regarding design choices

220021250- Tashfeea Alinur Rahman

- Designed the preliminary version of Snake and Ladder Position decoder
- Designed the Player Position Decoder
- Helped with writing of this report by providing a connection diagram

220021252- Yasin Hasan Saad

- Designed the final board module PCB by changing the placements and redrawing the traces
- Provided suggestions from time to time regarding design choices
- Wrote this report

CONCLUSION

This concludes the Hardware design portion of the project for EEE4308 course. In this phase, we had to rely Proteus (Ares) to design all the PCBs.

We plan to implement these designs in hardware if we get the opportunity further down the line.